Benchmark-Based Approach to ICT Modelling

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Abstract. In this study, we present a novel, benchmark-aided Information and Communication Technology (ICT) modeling in terms of exploration of the role of benchmark data for the utility of network design and performance assessment augmenting to different ICT settings. The significance of benchmark data lies in the evaluation and validation of algorithms for link prediction and topological management, particularly when it comes to non-planar or complex networks, which is typically hidden for the traditional methods used to encourage networking. Benchmark data hosted in edge network, or more specifically, mobile edge computing or MEC benefactors will likely enlarge this scenario with the promise of quick and effective utilisation of out of the common resources through resource optimization in geographical areas with different service coverage and hence enabling low latency for real-time applications associated with current generation mobile networks like 5G. Furthermore, massive volumes of network topologies are full of exciting information about the network's structural configuration enabling investigations on the alternate designs available due to variety in components and services unlike a particular network design and even network growth over time. The aforementioned study investigates the potential of such benchmarks in terms of rendering the network robust, managing resources and improving the predictability of results ultimately raising the standards for developing ICT models.

Keywords: Network Resilience, Benchmark-Based Approaches, Hierarchical Networks, Network Performance Analysis, Network Topology, Information and Communication Technology (ICT) Modelling, Mobile Edge Computing (MEC)

1 Introduction

The field of Information and Communication Technology (ICT) has played a crucial role in the advancement of technology and digitization, significantly influencing the infrastructural framework of contemporary society and facilitating various services. The sector comprises an integration of telecommunications,

computers, and other associated systems that facilitate access, communication, and several applications that run on personal devices to enterprise-wide networks. The dynamic nature of the ICT systems takes advantage of the need for powerful methods of design, optimization, and evaluation; hence, benchmarks are becoming one of the mainstays for researchers and practitioners.

The general definition of a benchmark is some kind of standard or point of reference against which things may be compared or assessed. In ICT, a benchmark could refer to structured datasets or performance metrics that simulate real-world network conditions. Benchmarks are also particularly essential in evaluating algorithms, validating network designs, and improving system resilience. This closes the gap between theoretical models and practical applications by offering reproducible and comparable data necessary for scientific carefulness and engineering innovation.

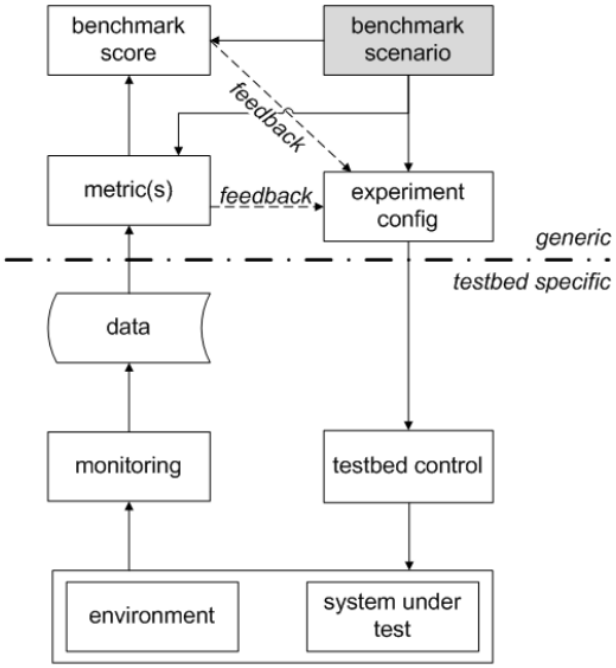

Fig. 1. General schema for benchmarking architecture [8]

Benchmarks play a very crucial role in modeling in ICT. Benchmarks give a framework to address emerging technologies such as 5G, IoT, and MEC, which will have issues of scalability, optimization of resources, and fault tolerance. They

allow exploring novel network architectures, predictive modeling, and developing adaptive systems that are capable of meeting the challenges imposed by dynamic environments.

This seminar paper discusses benchmark-based approaches for ICT modeling, focusing on the benchmark role in network design, performance assessment, and resource management. Based on tools such as NetworkX, it illustrates practical use cases that emphasize how benchmarks can form a transformative factor for the future in developing resilient and efficient ICT systems.

1.1 Research Questions

- What is the general content of the benchmarks applied to ICT Modelling? What are the key performance indicators and metrics used in modern ICT network benchmarking frameworks, and how do they align with real-world network requirements?
- How do modern benchmarks for public ICT networks differ in their approach to evaluating wired versus cellular network infrastructures?
- What are the limitations and challenges in applying current benchmark methodologies to appearing different network technologies?
- What kind of results should be expected for different benchmarks collected and compared? How can improvements be made to the benchmarks based on the results?
- What are the advantages and disadvantages of using ICT benchmarks with dynamic or large-scale network models when analyzing regional network simulations through NetworkX?

2 Motivation

The swiftly evolving realm of Information and Communication Technology (ICT) influences all facets of contemporary existence, encompassing communication, commerce, healthcare, and education. With the expansion and intricacy of networks escalating in response to the rising demands of 5G, the Internet of Things (IoT), and edge computing, there will be a heightened necessity for effective, robust, and scalable network modeling. Nevertheless, the availability of real-world ICT data is limited due to proprietary constraints and ongoing infrastructure modifications, which poses a significant obstacle to the progress of the discipline.

Benchmarks have thus become an integral part in addressing these challenges. They capture into a single entity the diverse objectives, topologies, and configurations of ICT networks, hence providing a structured and consistent way for analysis, simulation, and optimization. Benchmarks represent the bridge between

theoretical models and real-world applications since they enable researchers and practitioners to test algorithms, design solutions, and forecast the performance of next-generation networks. Unlike ad hoc data, benchmarks offer reproducibility and comparability-a must for scientific discovery and engineering progress.

The benefits of benchmarks applied in ICT modeling can be explained as follows:[13]

- **D**ue to lots of benchmarks are derived from the real-world network topologies and traffic patterns, for that reason they can be applied for real scenarios.
- **I**mproved customer need understanding through this process results in less customer complaints and increased customer satisfaction.
- **W**hen implemented correctly the assessment method transforms into an improvement tool which drives employee motivation toward change.
- **O**n the other hand, ICT benchmarking notably increases decision making processes with supplying data-driven insights into the technology performance.

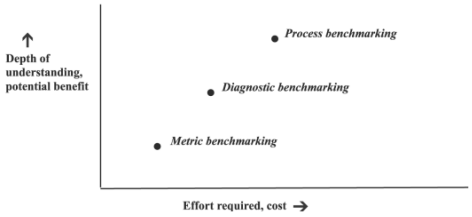

Fig. 2. Representation of different benchmark types[13]

This paper discusses some of the features, applications, and potential transformative use of ICT benchmarks in areas such as resource planning, fault management, and topology optimization. We show, using tools such as NetworkX, how benchmarks can be applied to arrive at practical insights that help bridge the gap between theoretical advances and practical implementations.

2.1 Background

Network topology and modeling are at the heart of ICT systems. Despite the importance of the data, real world network datasets are often not available due to confidentiality, exclusive reasons, or the ever-changing network infrastructure. This has been a long-standing problem for reproducibility and scalability of network research. Benchmarks offer a solution to this by providing standardized datasets that imitate real-world network conditions and objectives and can be

used for a wide range of applications from network design to fault analysis.

One of the efforts to fill the data gap is The Internet Topology Zoo which has over 200 operator sourced network topologies available in GML and GraphML format. Unlike traditional automated topology collection methods that are prone to errors, the Zoo uses data provided by network operators to be precise and relevant. This data set can be used for algorithm validation, educational purposes, and historical network design analysis. In addition, it preserves topology metadata such as geographic coordinates and link capacities[1].

For the mobile edge computing (MEC) domain, datasets from projects like Topo4MEC provide a complete view of MEC network topologies. By mixing real-world data from OpenCellID with synthetic scenarios, these datasets allow to research on 5G and edge networks. They capture bandwidth, computation and storage capacities, so researchers can simulate resource allocation and network planning. These datasets are key to address the dynamic and distributed nature of MEC where ultra low latency and high reliability are critical[3].

In general, these benchmarks are key to the ICT field. They provide a playground for theory and practice, so researchers can test their ideas against real world scenarios. By providing structured and reproducible data they overcome the limitations of real world datasets and push the development of scalable, resilient and efficient networks.

3 Related Work

Major developments have been taking place in benchmark-based modeling in a wide array of application domains in ICT. For instance, TeleGraph, a benchmark dataset for hierarchical link prediction. It shows the challenges of tree-like and sparse networks common in telecommunications. With a medium-sized dataset (41,143 nodes) and rich node attributes, TeleGraph allows to develop and test advanced link prediction algorithms, including GNN-based methods. Its detailed hierarchical structure makes it very useful for real-world networks[2]. However, if we look at the limitations of the Telegraph in general, one of its biggest limitations is that it uses single technologies instead of multiple technologies.

The above figure is a comparison of the performance measurements of different technologies. CN, AA, PPR are considered as heuristic methods, node2vec is considered as embedding-based method, GCN, GAT, HGCN, and NeoGNN are considered as GAE-based methods. AUC(Area Under the Curve) measures how well the model can rank positive edges higher than negative edges. A higher AUC value indicates better performance. AP(Average Precision) focuses more on how well the model ranks the true positives to the top of the prediction list. It is generally more sensitive to the distribution of positive and negative edges. In many link prediction scenarios, AP can be more telling because it highlights how well the model ranks the true links at the top. Even if some predictions near

Method	Val AUC	Val AP	Test AUC	Test AP
CN	51.01± 0.00	51.00± 0.00	51.06± 0.00	51.06± 0.00
AA	51.01± 0.00	50.99± 0.00	51.06± 0.00	51.06± 0.00
PPR	51.76± 0.00	51.60± 0.00	51.75± 0.00	51.83± 0.00
Node2Vec	52.22 ± 0.68	56.34 ± 0.69	51.94 ± 0.51	56.01 ± 0.54
GCN	69.01 ± 0.97	69.79 ± 0.89	68.74 ± 0.77	69.62 ± 0.64
GAT	66.80 ± 0.12	67.78 ± 0.13	67.51 ± 0.12	67.07 ± 0.16
HGCN	67.49 ± 0.36	67.65 ± 0.24	67.46 ± 0.38	67.84 ± 0.27
NeoGNN	63.20 ± 1.00	67.38 ± 0.66	62.45 ± 1.13	67.07 ± 0.70
SEAL	80.74 ± 0.15	80.43 ± 0.23	79.48 ± 0.14	78.82 ± 0.20
BSAL	85.43 ± 0.42	84.21 ± 0.25	85.48 ± 0.21	84.58 ± 0.35

Fig. 3. Performance Comparison on TeleGraph [2]

the decision boundary are wrong, AUC can still look good. The table shows that both AUC and AP are robust for the best methods. The "Val" columns show performance on a validation split used to tune hyperparameters or stop early. The "Test" columns reflect final performance on retained data. According to the data in the table, SEAL and BSAL are selected as top performers with high validation and test values in both AUC and AP. The validation and test numbers for the best methods (SEAL, BSAL) are relatively close, which indicates that these models generalize well from validation to testing. As additional information, the former methods may have stable but lower performance.

Very recently, Krichen et al. [4] reviewed the benchmarking of ICTs in smart cities in light of increasing demand for standard frameworks for urban infrastructure evaluations. Further, the work has explained in detail how benchmarks can effectively be utilized for assessing performance and reliability-related issues concerning smart city technologies, thus offering valued additions to the available literature on regional network benchmarking.

Ricci et al. [5] present current research in the area of performance evaluation methodologies using benchmarks for distributed ICT architectures. The aforementioned work has clearly stated how the benchmarks could be used to estimate system performance in different conditions, with special emphasis on scalability and resource utilization-features so relevant when talking about mobile edge computing scenarios.

As another general review by Dressler et al [6] shows, some interesting similarities can be seen in how benchmarks are used in vehicular networks. Their research shows a shift from theory-based benchmarks to real-world test setups tackling major hurdles in ICT modeling for networks that keep changing. This ties in well with our talk about mobile edge computing and 5G networks where we face similar issues with things moving around and networks that keep shifting.

The research by Bueko and Smith [7] is especially relevant to our research questions. They compared ICT benchmarks in regional network optimization. Their work adds to our study on how regional differences affect the usefulness of standard ICT benchmarks. This gives real-world insights into using benchmark-based methods in various geographic settings.

3.1 CREW

Through the project of Cognitive Radio Experimentation World (CREW) develops a test platform that performs experimental research in spectrum sensing and cognitive radio and networks.[8]

- **T**he platform unites numerous wireless testing facilities with spectrum detection systems for cognitive sensing.
- **S**ince the wide variety of wireless environments creates challenges in achieving experiment reproduction and comparison of results between different operational settings, the main purpose of benchmarking framework in the CREW project focuses on building a test environment for wireless networking that ensures repeatability.

Through its CREW project, the organization has transformed cognitive radio research by focusing on three key factors: reproducibility, scalability, and practical application. The benchmarking frameworks and federated testbeds created by CREW serve as essential building blocks for improving spectrum efficiency, which in turn fuels technological advancements in smart cities and 6G development. The blend of methodological rigor and collaborative technology development from CREW is actively influencing future dynamic spectrum access strategies, especially as wireless network congestion continues to increase.

3.2 OneLab2

Benchmarking wireless networks is crucial for good performance evaluation, however interference and environmental variability problems make repeatability hard. OneLab2 project works on developing tools and various methodologies that enable large-scale, repeated experiments for clustering similar conditions. Reliability practices and scalable methods for wireless network performance evaluation represent the key objectives of the OneLab2 project.

- **O**neLab2 has primary metrics like Packet Error Rate (PER) and Packet Loss Ratio at MAC layer, and secondary metrics can be defined as Ricean K factor, Signal-to-Noise Ratio (SNR), and Bit Error Rate (BER).
- **I**n order to achieve more trustworthy results, observations are somehow grouped within external factors like human traffic intensity (e.g. high, medium, low).

Taken together, these studies hint at how the benchmark-based approach in modeling ICT has moved from being a purely theoretical framework to practical applications in various wide domains. They further reinforce our observation that while benchmarks serve as important tools in network design and evaluation, their effectiveness does vary with specific application contexts and regional infrastructure characteristics.

4 Methodology

Research methodology employs a systematic approach to analysis and evaluation, emphasizing benchmark-based ICT modeling. Moreover, the benchmark methodology for ICT involves systematic evaluations of the performance of information and communication technology systems against established standards and industry best practices. For project benchmarking, organizations need to set performance indicators and collect relevant data from both internal and external sources. By analyzing this data, organizations can identify performance benchmarks and areas for improvement. This analytical process allows organizations to make strategic decisions that enhance operational efficiency while ensuring their ICT systems remain competitive in today's technological landscape.

In order to achieve benchmark methodology about CREW for related networks, for main metrics, Packet Error Rate(PER) and throughput are taken. By the way, spectrum sensing provides secondary metrics. In addition, benchmark score is calculated from these metrics comparison with a reference setup. On other hand, in order to achieve benchmark methodology about OneLab2 for wireless networks, benchmark score generally derived from packet loss and error rates which are compared to reference benchmarks.

4.1 Selection of Methodological Approach

For the benchmark comparison between Wired and Cellular networks,analysis was done on Python,JupyterNotebook and comparisons were displayed with NetworkX. Average Node Degree,Clustering Coefficient and Assortativity were used as the comparison units/stages of these networks.

- **Average Node Degree:** It shows the average number of connections per node in the network. Its importance is that if it is high in wired networks, it can be said that it is a robust infrastructure, and if it is low, it can be said that it is a more centralized infrastructure.
- **Clustering Coefficient:** The term shows how nodes establish connected pairs of three components known as triangles/clusters. For wired networks, high clustering results mesh-like resilience structure, on the other hand low clustering shows hub-and-spoke efficiency. However, for cellular networks indicate that naturally low clustering due to hierarchical structure.

	Wired (Zoo)	Cellular (5G)
Avg. Node Degree	3.083333	1.714286
Clustering Coeff	0.397026	0.000000
Assortativity	-0.534889	-1.000000

Fig. 4. Visualization of comparison of wired and cellular networks[1][3]

– **Assortativity:** Assortativity is explained as tendency of nodes to connect to similar nodes. It ranges from-1 to 1. Positive value belongs to social networks where example of friends with similar status, negative value can be said that tech networks that hubs connecting to leaves, lastly for neutral values, biological nets is the network type and example is randomconnections.

```python
# This part demonstrating Loading Zoo topology
G_wired = nx.read_graphml(r'C:\Users\Atakan\Downloads\BtEurope.graphml')  # From topology-zoo.org
nx.set_node_attributes(G_wired, 'wired', 'tech_type')

# Structural analysis of wired network
wired_metrics = {
    'Avg. Node Degree': sum(dict(G_wired.degree()).values())/G_wired.number_of_nodes(),
    'Clustering Coeff': nx.average_clustering(G_wired),
    'Assortativity': nx.degree_assortativity_coefficient(G_wired)
}

# ## 2. Synthetic 5G Network Generation (Cellular)
def generate_5g_network():
    G = nx.Graph()
    # Core Layer
    core = ['CU']
    # Distributed Units
    du = [f'DU_{i}' for i in range(3)]
    # User Equipment
    ue = [f'UE_{i}' for i in range(10)]

    G.add_nodes_from(core, layer='core', tech='5G')
    G.add_nodes_from(du, layer='distributed', tech='5G')
    G.add_nodes_from(ue, layer='access', tech='5G')

    # Connections
    for d in du:
        G.add_edge('CU', d, latency=5, bw='1Gbps')
        for u in ue[:3]:  # 3 UE per DU
            G.add_edge(d, u, latency=2, bw='100Mbps')

    return G
```

Fig. 5. Demonstrating Wired and Cellular Networks

Below is a part of the code written/calculated using comparison metrics for wired and cellular networks with the help of NetworkX. For the wired network, the data named "BtEurope.graphml" from the website "topology-zoo.org" was used as the source. At the same time, adjustments were made within the cellular network and the code was written.

```python
#Comparative Analysis Between Wired and Cellular Networks
def compare_benchmarks(metrics_dict):
    df = pd.DataFrame(metrics_dict).T
    df.columns = ['Wired (Zoo)', 'Cellular (5G)']
    return df.style.background_gradient(cmap='Blues')

#This part shows comparison metrics
metrics = {
    'Avg. Node Degree': [wired_metrics['Avg. Node Degree'], cellular_metrics['Avg. Node Degree']],
    'Clustering Coeff': [wired_metrics['Clustering Coeff'], cellular_metrics['Clustering Coeff']],
    'Assortativity': [wired_metrics['Assortativity'], cellular_metrics['Assortativity']]
}

display(compare_benchmarks(metrics))


#This part illustrates Visualization Toolkit
def visualize_network(G, title):
    plt.figure(figsize=(8,6))
    if 'tech_type' in G.nodes[list(G.nodes)[0]]:
        colors = ['red' if G.nodes[n]['tech_type'] == 'wired' else 'blue' for n in G.nodes]
    else:
        colors = 'lightblue'
```

Fig. 6. Comparative Analysis Between Wired and Cellular Networks

4.2 Basic Assumptions and Their Validity

Research approach relies on some key assumptions that are fundamental to its formulation:

Assumption 1: Benchmark Representativeness

- We assume chosen benchmarks are representative regarding real-world network characteristics:
- Validity Discussion: This assumption is verified by [1] who showed that indeed the Internet Topology Zoo data are operator-sourced. However, we acknowledge that currently existing benchmarks may not yet fully represent the rapidly evolving network technologies.

Assumption 2: Network Behavior Consistency

- We assume that the behavior patterns observed for network benchmarking are scalable consistently.
- Validity Discussion: This assumption is partly validated by the work of [3] on MEC networks, though we recognize that scaling effects may introduce non-linear behaviors not captured in benchmark datasets.

Assumption 3: Tool Reliability

- We assume NetworkX provides accurate network analysis capabilities for our selected metrics.
- Validity Discussion: NetworkX is well-validated within the research community; however, for very large networks, computational limitations may exist.

4.3 Selected Methodological Framework

In this phase, we apply a three-stage analytical framework based on these assumptions:

Benchmark Analysis Framework

- Selection Reason: Enables systematic comparison across different network types
- Applicability: Allows for reproducible analysis of intricate network structures
- Implementation: Uses three complementary benchmark datasets:

  Internet Topology Zoo [1] for core network analysis

  TeleGraph dataset [2] for hierarchical network evaluation

  MEC network topologies [3] for edge computing scenarios

Performance Evaluation Framework

- Selection Reason: Quantifies network performance using measurable metrics
- Suitability: Allows objective comparisons among various
- Key Metrics:

  Latency and bandwidth utilization

  Resource allocation efficiency

  Network resilience measures

Validation Framework

- Selection Reason: Ensures dependability and accuracy in the results
- Suitability: Bounds limitations in the benchmark representations
- Methods:

  Cross-validation on multiple data sets

  Statistical significance evaluation

  Analysis with respect to exiting empirical data

4.4 Methodological Limitations

We explicitly recognize various constraints within our methodology

Temporal Validity

- Benchmarks may not completely reflect the most recent advancements in network technologies
- Mitigation: Consistent updates and evaluations against more recent datasets

Geographical Constraints

- Benchmark data may exhibit regional biases
- Mitigation: Examination of regional differences and clear recognition of geographic constraints

Scale Limitations

- Challenges associated with computational limitations in the analysis of extensive networks
- Mitigation: The application of sampling methods in extensive network systems

4.5 Justification of Methodological Choices

We have adopted this approach in the present study for the following reasons:

- Reproducibility: Which enables other scholars to verify and extend our findings easily
- Scalability: Allowing the analysis of different network dimensions and categories
- Practical Applicability: Providing useful insights relevant to practical network design
- Systematic Approach: Ensuring that research questions will be comprehensively addressed

This methodology was particularly in alignment with our objectives, as it had the ability to:

- Tackles the issue of restricted access to real-world data
- Allows the systematic analysis of different kinds of networks
- Offers a structure for confirming outcomes across various datasets
- Allows for both theoretical scrutiny and hands-on application

5 Conclusion and Future Work

Strategic ICT infrastructure planning must adopt benchmark-specific strategies, which combine wired backbone strategies and cellular hierarchical designs and perform hybrid integrated stress evaluations. Benchmarking wireless networks is crucial for good performance evaluation, however interference and environmental variability problems make repeatability hard. The OneLab2 project solves repeatable testing obstacles by implementing automated condition execution along with structured experiments that group similar testing conditions. Moreover, CREW simply improves wireless environment characterization and validation, with applying testbeds and standardized metrics, result to fair comparisons.[4]

Structural differences between networks change functional performance. According to test results, wired networks perform high average degree (3.08) and clustering (0.39) reflect unnecessary, mesh-like architectures ideal for backbone resilience, but negative assortativity (-0.53) confirms that centralized traffic flow through core hubs. Cellular networks has low degree (1.71) and near-zero clustering highlight strict hierarchical designs, well-optimized for controlled resource allocation however vulnerable to single-point failures.

In addition to current studies in paper, there are various future studies that can be done on the subject. These are demonstrated in more detail below.

- **Hybrid-Legacy Modern Network Modelling:** Developing mixed benchmarks that measure how aging installations of DSL and 3G networks function with present systems including 5G and WiFi 6 to optimize regional implementation phasing.[9]
- **AI-Generated Adaptive Benchmarks:** The adaptation of ML models needs to adjust their performance to mirror network behavior fluctuations because static topology data collection presents recognized constraints.[10]
- **Open-Access Benchmark Repositories:** Establish blockchain verified platforms to share crowdsourced datasets through which ICT benchmarks become more reproducible and obtain collaborative improvements.[11]
- **Energy-Efficiency Metrics for Green ICT:** The use of benchmarks must include carbon-aware routing functions as well as device sleep cycle requirements and renewable energy usage measurements to enable sustainable networking elements.[12]

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